

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5-ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5– Hackathon	Assessment:	Assessment Tool 3– Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning(BL4)		
Student Name:	Shaik Nuha	Reg. No.:	192472242
Section / Batch:	2024	Date:	31-08-2026

Code of Conduct

I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk.Nuha

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem Statement:

Energy Consumption Estimation. Predict a households or buildings electricity consumption based on temperature, occupancy, and appliance usage patterns.

Code:

```
Explorer
  FOD-COS-AT3
    energy_consumption_estimator.py
    energy_dataset.csv
    prediction_history.csv

Energy Consumption Estimator
  import os
  import csv
  import datetime
  import numpy as np
  import pandas as pd
  import tkinter as tk
  from tkinter import ttk, messagebox, filedialog
  import matplotlib
  matplotlib.use("TkAgg")
  import matplotlib.pyplot as plt
  from sklearn.linear_model import LinearRegression
  from sklearn.model_selection import train_test_split
  from sklearn.metrics import r2_score
  from sklearn.preprocessing import StandardScaler

  # ---
  # CONSTANTS & THEME
  # ---
  COST_PER_KWH = 8 # Rs. per kwh
  HISTORY_CSV = "prediction_history.csv"
  DATASET_CSV = "energy_dataset.csv"
  HISTORY_COLUMNS = [
    "Timestamp", "Temperature(C)", "Occupants",
    "ApplianceHours", "ACHours", "HeaterHours",
    "DayType", "Predicted_KWh", "EstimatedCost(Rs)"
  ]

  # Colour palette - deep navy + electric teal accent
  BG_DARK = "#001B2A"
  BG_MID = "#1B2B3B"
  BG_CARD = "#1E3448"
  ACCENT = "#00C9A7"
  ACCENT2 = "#00B0A8"
  TEXT_LIGHT = "#E8F4FD"
  TEXT_DIM = "#8BA7BF"
```

Output:

Energy Consumption Estimation System

Monday, 31 Aug 2026 08:45:18

Energy Consumption Estimation

Input Parameters

Temperature (C)

35

Number of Occupants

4

Appliance Usage (hrs)

6

AC Usage (hrs/day)

4

Heater Usage (hrs/day)

1

Day Type

Weekday

Model R2 Score: 0.9223

Predict

Clear

Exit

Prediction Result

Predicted Energy Consumption: 31.00 kWh

Estimated Monthly Cost: Rs.7440.00

Data Visualisations

Temperature vs Consumption

Occupancy vs Consumption

Feature Importance

Prediction History

View History

Export History CSV

Cost Rate

Rs.8/kWh

Algorithm

Linear Regression

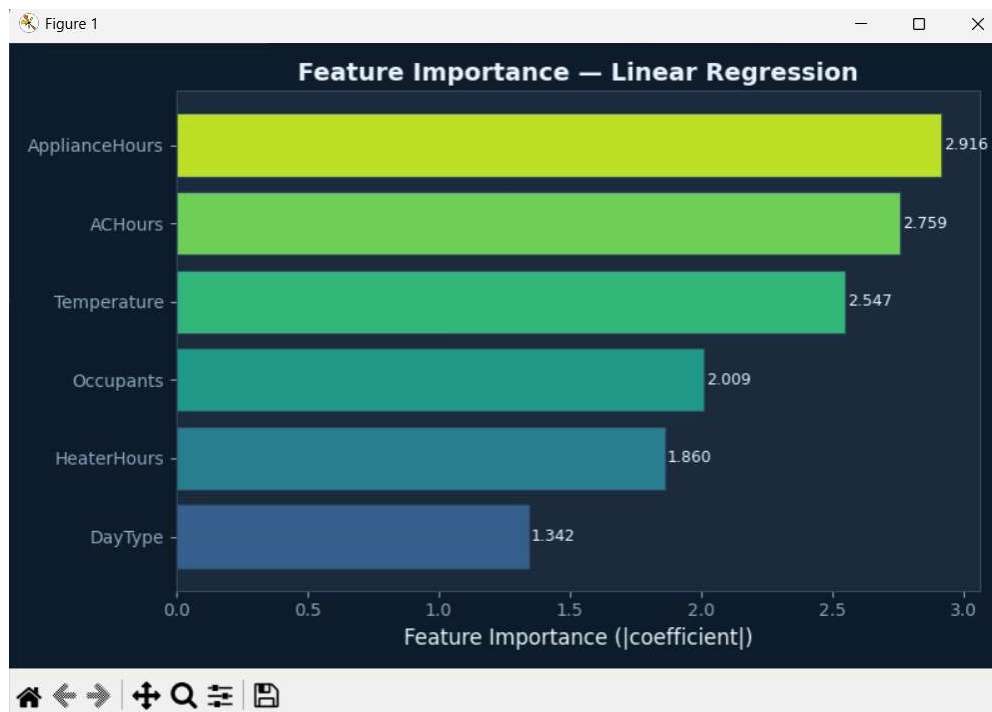
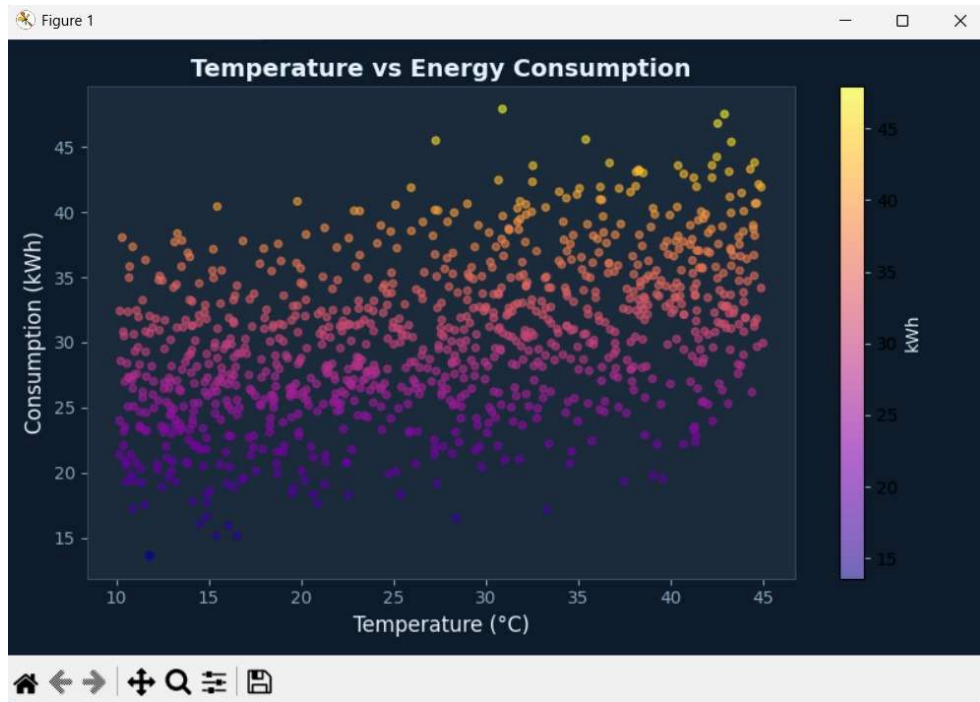
Train Split

80 / 20

Dataset

1 000 samples

Predictions: 31.00 kWh | Daily cost: Rs.248.00 | Monthly est: Rs.7440.00 | Saved to history



GEO TAG:

